

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA0306	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
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Section / Batch:	CSE AI	Date:	13-07-2026

Section 1: Regular Expression Pattern Generation

Code:

```
RE Pattern.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\RE Pattern.py (3.12.3)
File Edit Format Run Options Window Help

import re

text = """
Name: John Doe
Email: john.doe123@gmail.com
Mobile: +91-9876543210
Password: P@ssw0rd123
Date of Birth: 15/08/2004
Register Number: 23AIML1056
Department: Artificial Intelligence and Machine Learning
"""

email_pattern = r'[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)'
mobile_pattern = r'\+91-\d{10}'
password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#$$%&!*]) [A-Za-z\d@#$$%&!*]{8,}$'
dob_pattern = r'\d{2}/\d{2}/\d{4}'
reg_pattern = r'\d{2}[A-Z]{4}\d{4}'

print("Email:", re.search(email_pattern, text).group())
print("Mobile:", re.search(mobile_pattern, text).group())
print("DOB:", re.search(dob_pattern, text).group())
print("Register Number:", re.search(reg_pattern, text).group())

password = "P@ssw0rd123"

if re.match(password_pattern, password):
    print("Password: Valid")
else:
    print("Password: Invalid")
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\RE Pattern.py
Email: john.doe123@gmail.com
Mobile: +91-9876543210
DOB: 15/08/2004
Register Number: 23A1ML1056
Password: Valid
>>>
```

Section 2: Search for a String

Code:

```
String search.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\String search.py (3.12.3)
File Edit Format Run Options Window Help
import re

text = """Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud,
and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making.
As AI continues to evolve, professionals with AI skills are in high demand."""

word = "AI"

match = re.search(word, text)

if match:
    print("First Occurrence:", match.group())
    print("Starting Position:", match.start())
    print("Ending Position:", match.end())
    print("Total Occurrences:", len(re.findall(word, text)))
else:
    print("Word not found")
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\String search.py
First Occurrence: AI
Starting Position: 25
Ending Position: 27
Total Occurrences: 6
>>>
```

Section 3: Split the documentation into sentences and words

Code:

```
split.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\split.py (3.12.3)
File Edit Format Run Options Window Help

import re

text = """Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud,
and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making.
As AI continues to evolve, professionals with AI skills are in high demand!"""

sentences = [s.strip() for s in re.split(r'[.!?]+' , text) if s.strip()]

words = re.split(r'\s+', text)

print("Total Sentences:", len(sentences))
print("Total Words:", len(words))

print("\nSentences:")
for s in sentences:
    print(s)

print("\nWords:")
print(words)
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help

Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

>>> = RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\split.py
Total Sentences: 4
Total Words: 60

Sentences:
Artificial Intelligence (AI) is transforming industries across the world
AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud,
and in education to provide personalized learning experiences
Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making
As AI continues to evolve, professionals with AI skills are in high demand

Words:
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used', 'in', 'healthcare', 'to', 'assi
st', 'doctors', 'in', 'diagnosis,', 'in', 'banking', 'to', 'detect', 'fraud,', 'and', 'in', 'education', 'to', 'provide', 'personalized', 'learning', 'ex
periences.', 'Many', 'companies', 'invest', 'heavily', 'in', 'AI', 'research', 'because', 'AI', 'improves', 'efficiency', 'and', 'enables', 'intelligent'
, 'decision-making.', 'As', 'AI', 'continues', 'to', 'evolve,', 'professionals', 'with', 'AI', 'skills', 'are', 'in', 'high', 'demand!']

>>>
```

Section 4: Email and Strong Password Validation

Code:

```
email and pass.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\email and pass.py (3.12.3)
File Edit Format Run Options Window Help

import re

email = input("Enter Email: ")
password = input("Enter Password: ")

email_pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$'

password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#$$%&!*])[A-Za-z\d@#$$%&!*]{8,}$'

if re.match(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")

if re.match(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\email and pass.py
Enter Email: sravan12@gmail.com
Enter Password: harsha@434
Valid Email
Weak Password
>>>
```

Section 5: Compile Once, Validate Many

Code:

```
valid many.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\valid many.py (3.12.3)
File Edit Format Run Options Window Help
import re

emails = [
    "john@gmail.com",
    "alice@yahoo.org",
    "invalid@",
    "student123@college.edu",
    "abc@gmail"
]

pattern = re.compile(r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$')

for email in emails:
    if pattern.match(email):
        print(email, "-> Valid")
    else:
        print(email, "-> Invalid")
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\valid many.py
john@gmail.com -> Valid
alice@yahoo.org -> Valid
invalid@ -> Invalid
student123@college.edu -> Valid
abc@gmail -> Invalid
>>>
```